FLATNESS OF A MODULE OVER THE BASE LOCAL RING AND OVER THE LOCAL RING OF AN ETALE COVER ARE EQUIVALENT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Pushforward of a quasi-coherent sheaf along the canonical morphism from the spectrum of a local ring. Schemes, label schemes-lemma-push-forward-quasi-coherent, invoked at proof line 174. It shows that $\mathcal{F} = (\Spec(\mathcal{O}_{T,t}) \to T)_*\widetilde M$ is a quasi-coherent sheaf on T, so that the affine form of the criterion can be applied to it.
- The affine form of the criterion for an etale ring map. Same chapter flat.tex, label lemma-etale-flat-up-down, invoked at proof line 179. Applied to the sheaf whose stalk at t is M, it gives the equivalence of flatness over $\mathcal{O}_{S,s}$ and over $\mathcal{O}_{T,t}$.

Source reference. Original lemma 05B9: sources/stacks-flat/pinned-original-source.tex, source lines 124–124. Navigation pointer for \ref{lemma-etale-flat-up-down} used inside the counted span (line 179); the argument that consumes it is retained in the span.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Navigation pointer for \ref{schemes-lemma-push-forward-quasi-coherent} used inside the counted span (line 175); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $T \to S$ be an étale morphism. Let $t \in T$ with image $s \in S$. Let M be a $\mathcal{O}_{T,t}$-module. Then*

$$M \text{ flat over } \mathcal{O}_{S,s} \Leftrightarrow M \text{ flat over } \mathcal{O}_{T,t}.$$

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Proof. We may replace S by an affine neighbourhood of s and after that T by an affine neighbourhood of t. Set $\mathcal{F} = (\mathrm{Spec}(\mathcal{O}_{T,t}) \to T)_*\widetilde{M}$. This is a quasi-coherent sheaf (see Schemes, Lemma 01LC or argue directly) on T whose stalk at t is M (details omitted). Apply Lemma 05B9. $\square$